# PREPROCESSING TO PREDICTION

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# OUTLINE

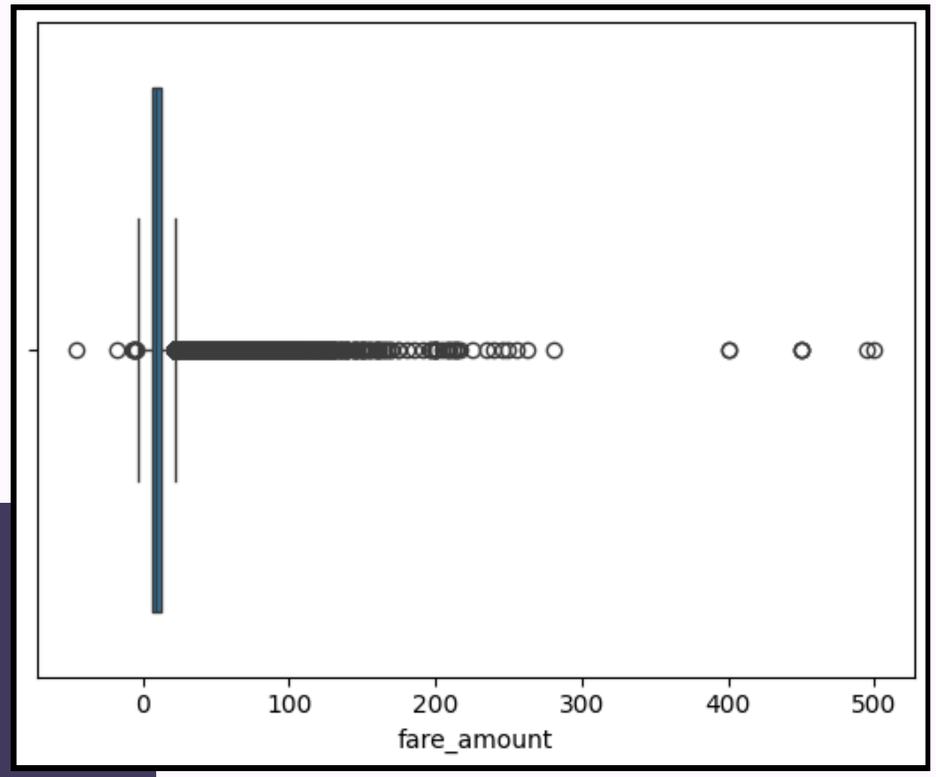
- Data Preprocessing
- Feature Engineering
- Feature Selection
- Feature Scaling
- Encoding
- Model Selection
- Evaluation & Optimization

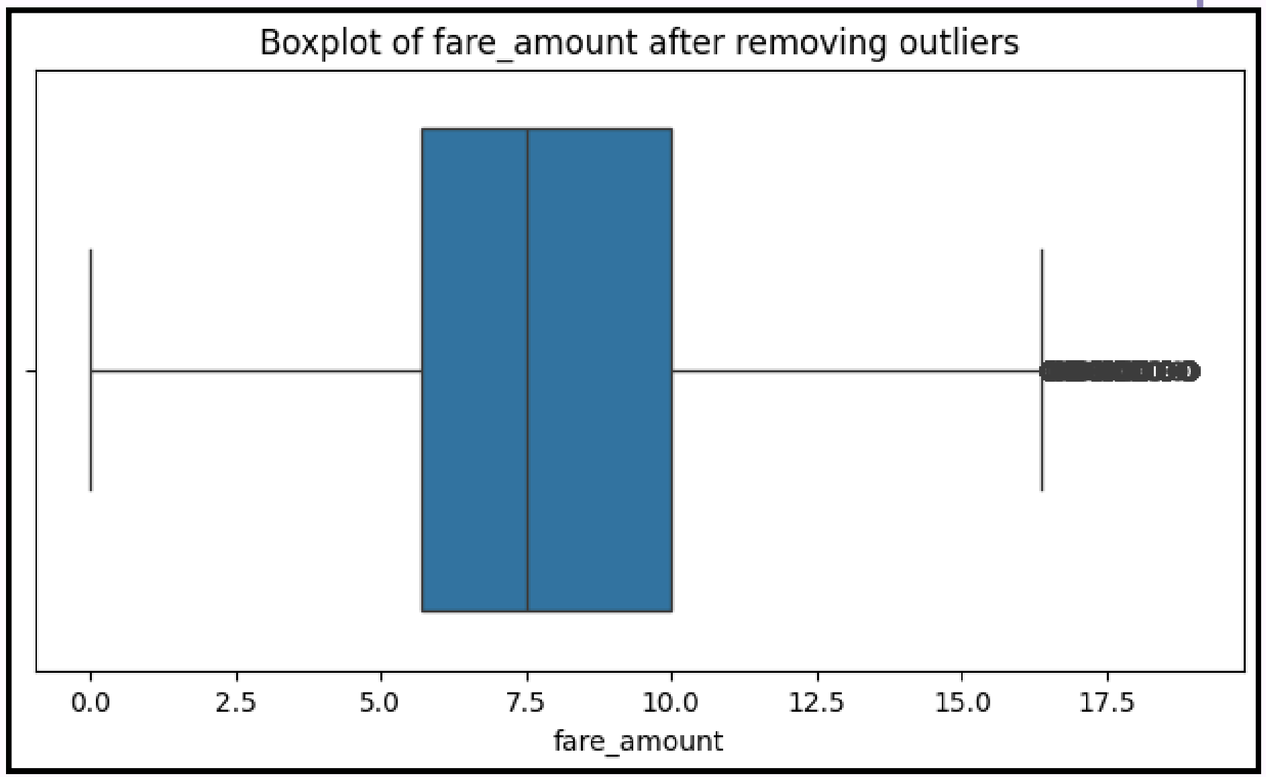
# INTRODUCTION

- The dataset contains 26 features describing Ride Context, Location and time information with 500,000 rows.
- Our target variable is the fare amount of each ride.

# PREPROCESSING

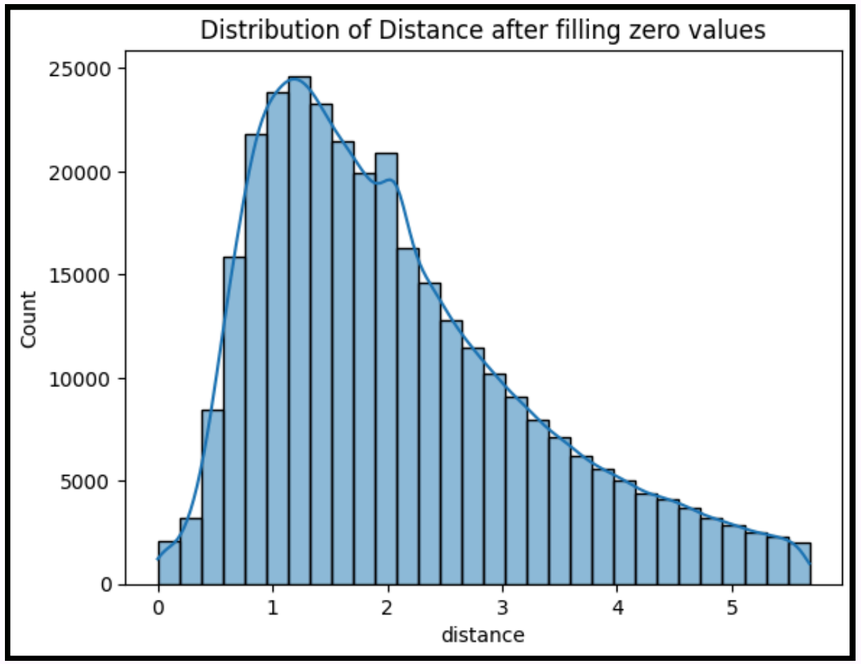
- Handling Outliers



# PREPROCESSING

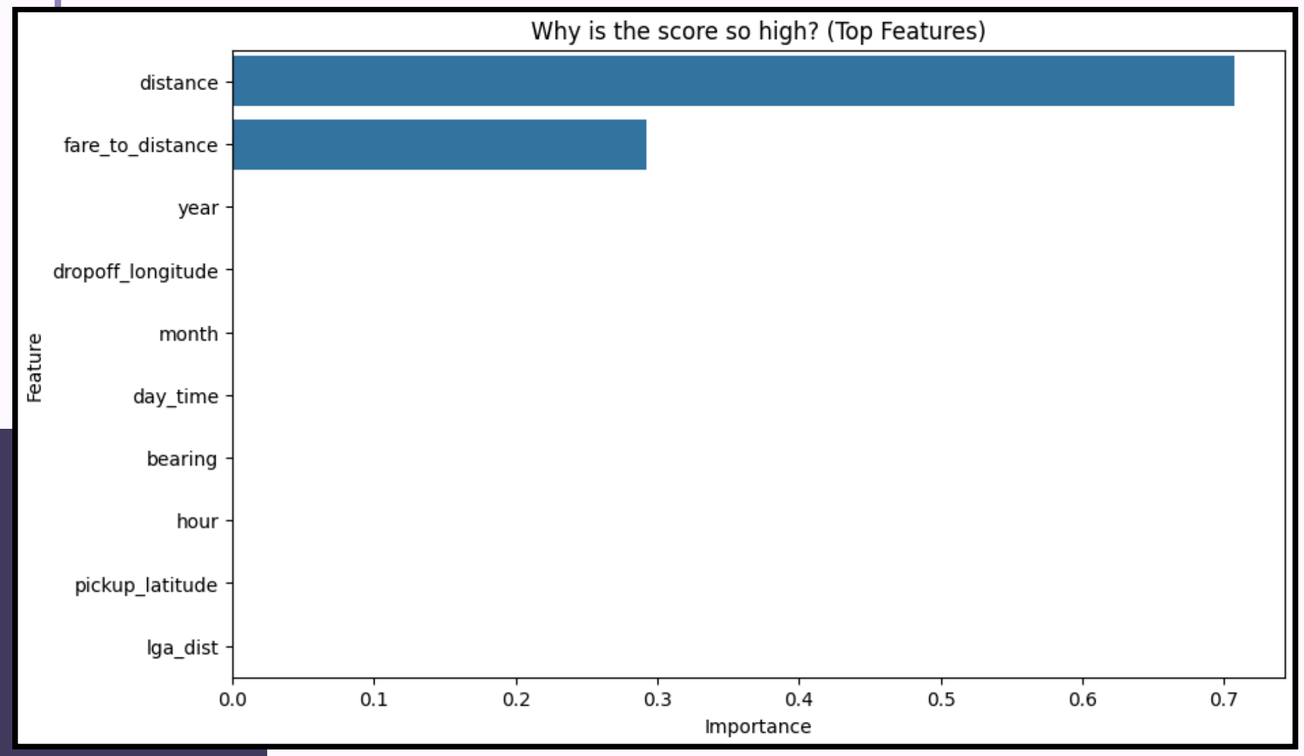
- Checking for missing values
- Removing/filling missing values
- Checking for duplicates

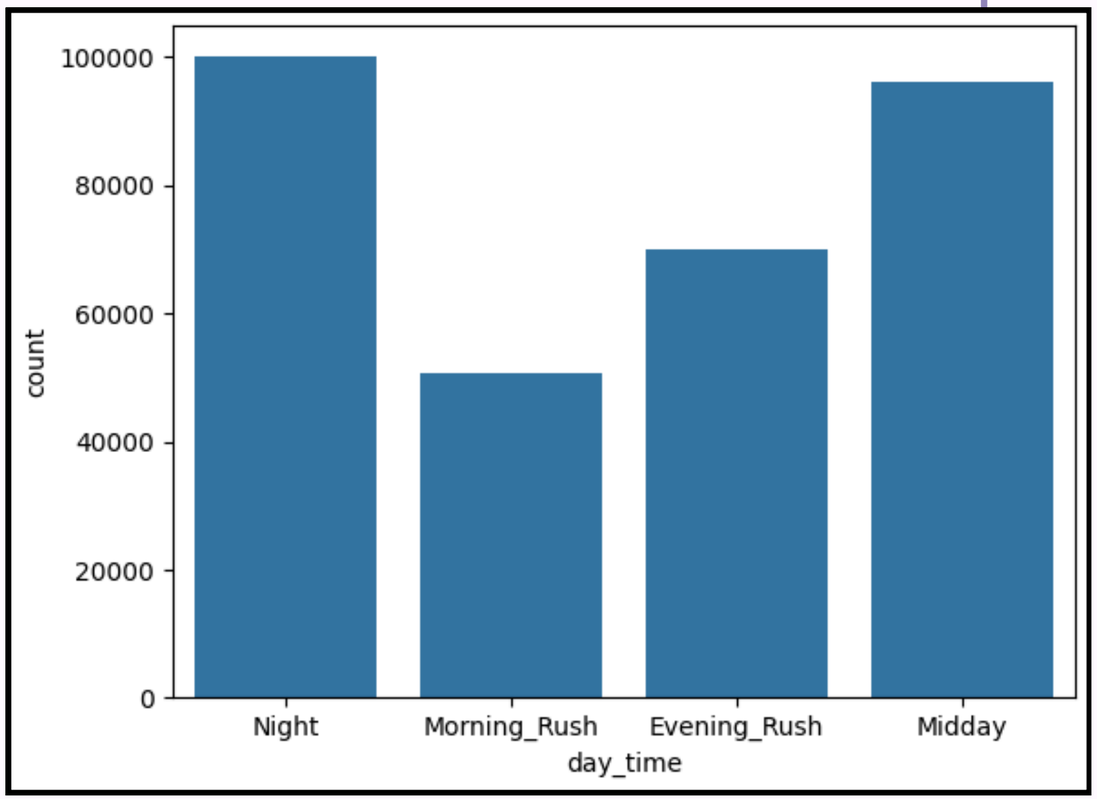



| | |
|---|---|
| User ID | 0 |
| User Name | 0 |
| Driver Name | 0 |
| Car Condition | 0 |
| Weather | 0 |
| Traffic Condition | 0 |
| key | 0 |
| fare_amount | 0 |
| pickup_datetime | 0 |
| pickup_longitude | 0 |
| pickup_latitude | 0 |
| dropoff_longitude | 5 |
| dropoff_latitude | 5 |
| passenger_count | 0 |
| hour | 0 |
| day | 0 |
| month | 0 |
| weekday | 0 |
| year | 0 |
| jfk_dist | 5 |
| ewr_dist | 5 |
| lga_dist | 5 |
| sol_dist | 5 |
| nyc_dist | 5 |
| distance | 5 |

# FEATURE ENGINEERING

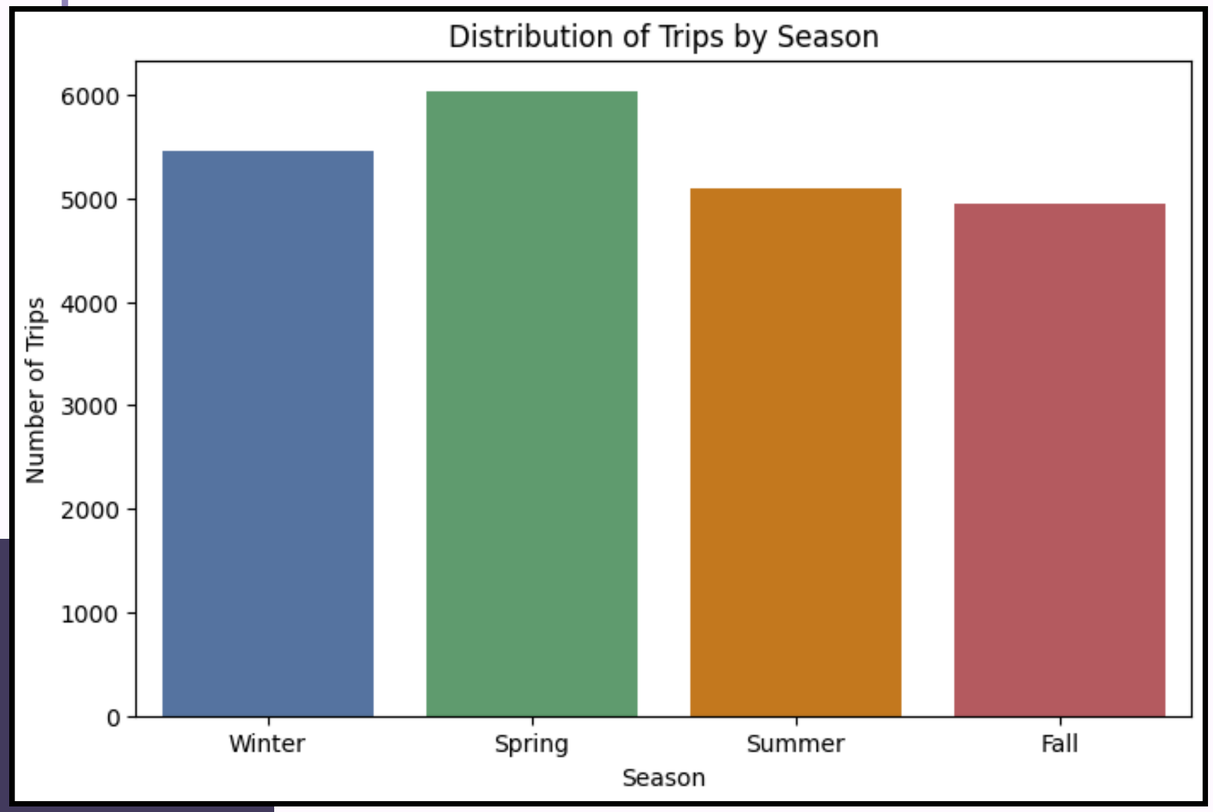
- Creating new feature (day_time)
- Creating new feature (fare_to_distance), then removing it after it lead to model overfitting.

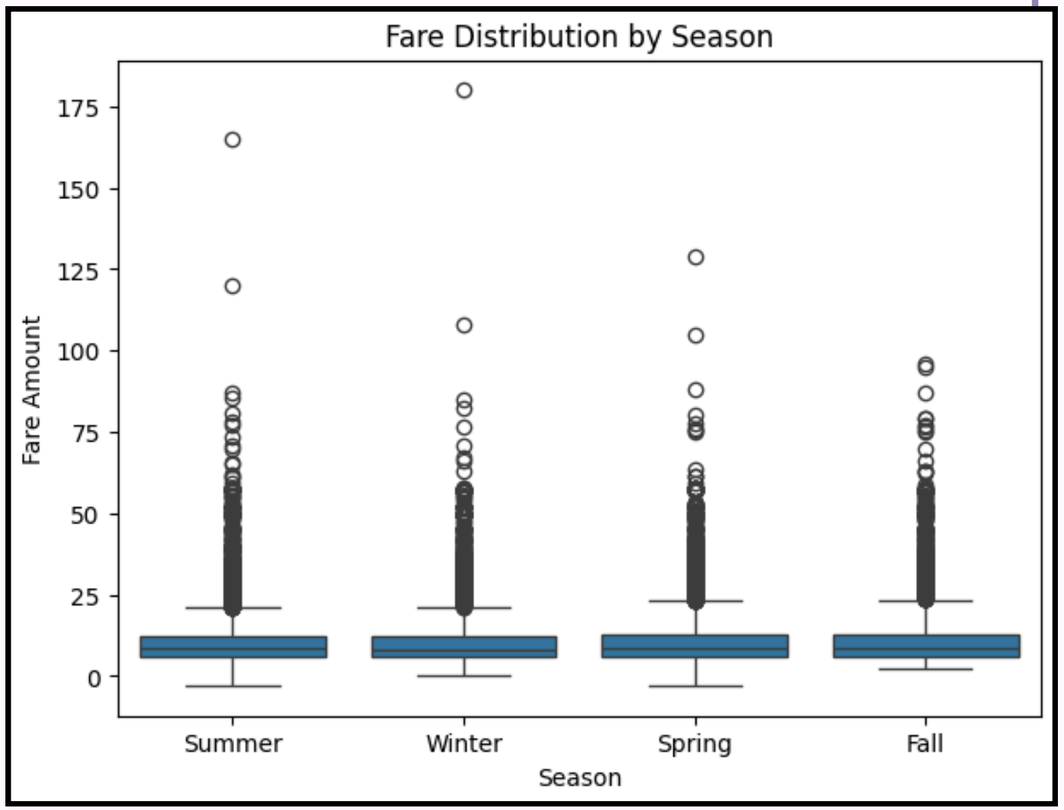


# FEATURE ENGINEERING

- Creating new feature of distributing trips by season



# FEATURE SELECTION

- Dropping redundant variables
- Dropping unnecessary variables

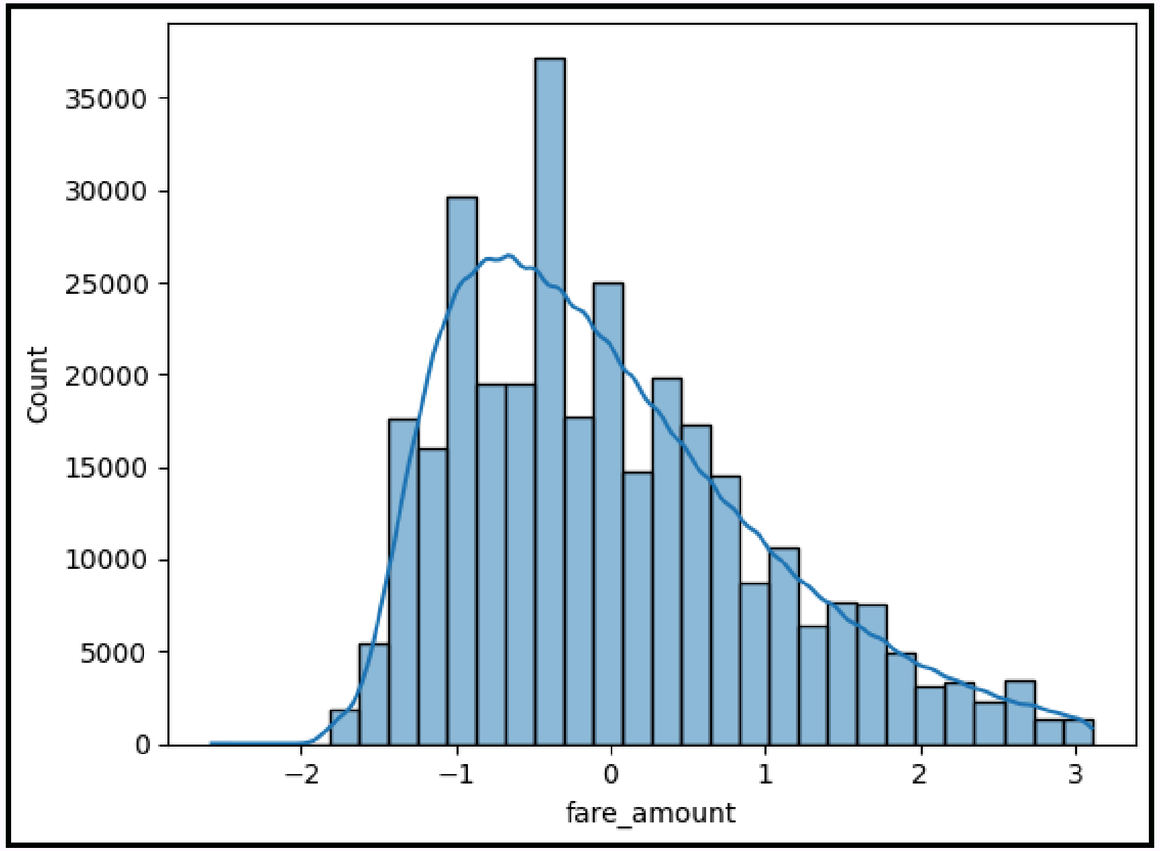
User ID
User Name
Driver Name

key

| pickup_datetime |
| --- |
| 2009-06-15 17:26:21 |
| 2010-01-05 16:52:16 |
| 2011-08-18 00:35:00 |
| 2012-04-21 04:30:42 |
| 2010-03-09 07:51:00 |

# FEATURE SCALING

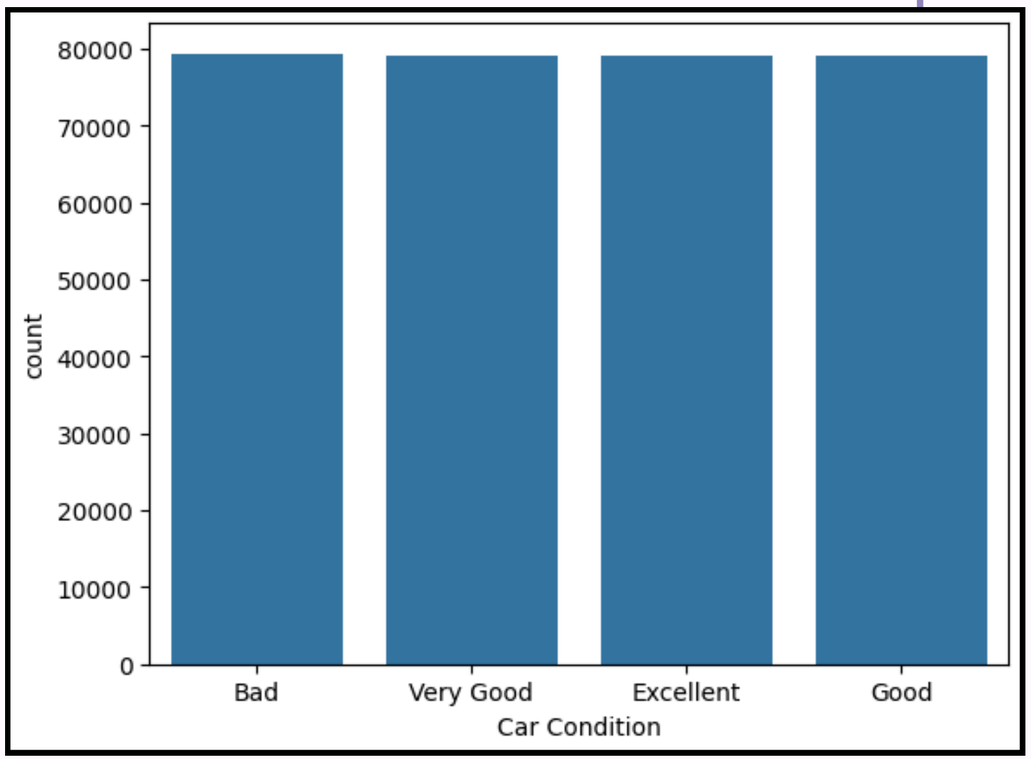
- Standardization for numerical values



# ENCODING

- Label Encoding for Ordinal data
- One-Hot encoding for nominal data

| | Weather_cloudy | Weather_rainy | Weather_stormy | Weather_sunny | Weather_windy |
|---|---|---|---|---|---|
| 2 | 0 | 0 | 1 | 0 | 0 |
| 3 | 0 | 0 | 1 | 0 | 0 |
| 4 | 0 | 0 | 1 | 0 | 0 |
| 5 | 0 | 0 | 0 | 1 | 0 |
| 6 | 0 | 0 | 0 | 0 | 1 |



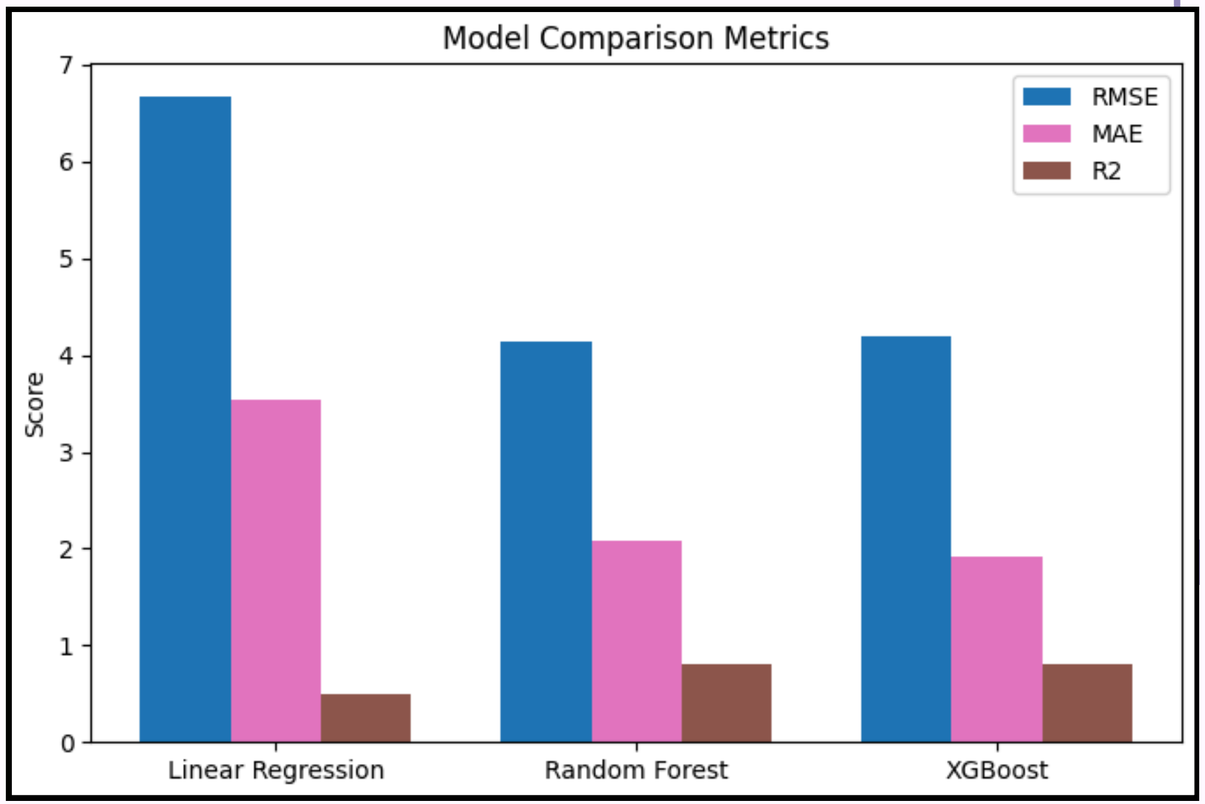
# MODEL SELECTION

- Linear Regression
- Random Forest
- XGBoost
- Decision Tree

| Model | RMSE | MAE | R2 |
|---|---|---|---|
| Linear Regression | 6.677159 | 3.532748 | 0.495550 |
| Random Forest | 4.133599 | 2.076383 | 0.806673 |
| XGBoost | 4.197390 | 1.915638 | 0.800660 |

# EVALUATION

- Random Forest model showed the highest score among the models, with R2 score =0.8



# THANK YOU